Chapter 12: Differentiation

Chapter 12 · Original slide 1 · Differentiation

Pure Y1 Ch 12
Differentiation

Chapter 12 · Original slide 2 · Chapter Overview

Those who have done either IGCSE Mathematics, IGCSE Further Mathematics or Additional Mathematics would have encountered this content. Otherwise it will be completely new!

1: Find the derivative of polynomials.

2: Find equations of tangents and normal to curves.

If $y = 3x^2 + \sqrt{x}$, find $\frac{dy}{dx}$

The point $P\left(3, 9\right)$ lies on the curve C with equation $y = x^2$. Determine the equation of the tangent to C at the point P.

3: Identify increasing and decreasing functions.

Find the range of values for which $f\left(x\right) = x^3 - x$ is increasing.

4: Find and understand the second derivative $\frac{d^2y}{dx^2}$ or $f''\left(x\right)$

If $y = x^4 - 3x^2$ determine $\frac{d^2y}{dx^2}$

5: Find stationary points and determine their nature.

6: Sketch a gradient function.

Find the stationary points of $y = x^3 - x$ and state whether each is a maximum or minimum point.

Draw $y = x^3$ and its gradient function on the same axes.

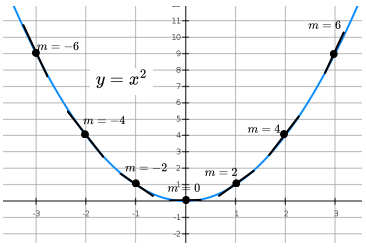
7: Model real-life problems.

Chapter 12: Differentiation

Chapter 12 · Original slide 3 · Gradient Function



$m = 3$



At GCSE, you found the gradient of a curve at a particular point by drawing a tangent.

For a straight line, the gradient is constant:

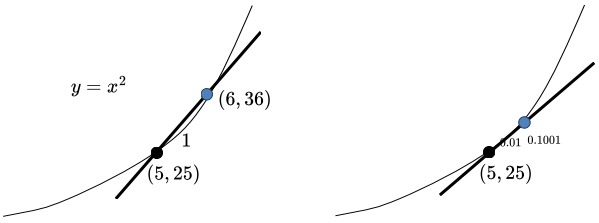
$y = 3x + 2$

However, for a curve the gradient varies. We can no longer have a single value for the gradient; we ideally want an expression in terms of $\mathbf{x}$ that gives us the gradient for any value of x (unsurprisingly known as the gradient function).

By looking at the relationship between x and the gradient at that point, can you come up with an expression, in terms of x for the gradient?

Gradient Function $=$

Chapter 12 · Original slide 4 · Finding the Gradient Function



The question is then: Is there a method to work out the gradient function without having to draw lots of tangents and hoping that we can spot the rule?

To approximate the gradient on the curve $y = x^2$ when $x = 5$, we could pick a point on the curve just slightly to the right, then find the gradient between the two points:

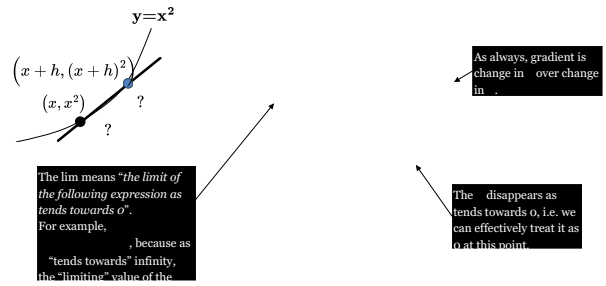
$(5.01, 25.1001)$

As the second point gets closer and closer, the gradient becomes a better approximation of the true gradient:

The actual gradient when $x = 5$ is 10, so this approximation is damn close!

Chapter 12: Differentiation

Chapter 12 · Original slide 5 · Finding the Gradient Function



This gives us a numerical method to get the gradient at a particular **x**, but doesn't give us the gradient function in general. Let's use exactly the same method, but keep x general, and make the 'small change' (which was previously 0.01) 'h':

And voila, we got the $2x$ we saw earlier!

Chapter 12 · Original slide 6 · Finding the Gradient Function

! The gradient function, or derivative, of the curve $y = f(x)$ is written as $f\prime(x)$ or $\frac{dy}{dx}$.

$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$

The gradient function can be used to find the gradient of the curve for any value of x.

We will soon see an easier/quicker way to differentiate expressions like $y = x^3 - x$ without using 'limits'. But this method, known as differentiating by first principles, is now in the A Level syllabus, and you could be tested on it!

Notation Note:

Whether we use $\frac{dy}{dx}$ or $f\prime(x)$ for the gradient function depends on whether we use $y =$ or $f(x) =$ to start with:

$y = x^2 \to \frac{dy}{dx} = 2x$
$f(x) = x^2 \to f'(x) = 2x$

"Leibniz's notation"

"Lagrange's notation"

Advanced Notation Note:

Rather than h for the small change in x, the formal notation is δx. So actually:

$f'(x) = \lim_{\delta x \to 0} \frac{f(x+\delta x) - f(x)}{\delta x}$

So we in fact have 3 symbols for "change in"!

- Δx: any change in x (as seen in Chp5: $m = \frac{\Delta y}{\Delta x}$)
- δx: a small change in x
- dx: an infinitesimally small change in x

Chapter 12 · Original slide 7 · Example

The point A with coordinates $(4, 16)$ lies on the curve with equation $y = x^2$.

At point A the curve has gradient g.

a) Show that $g = \lim_{h \to 0} (8 + h)$

b) Deduce the value of g.

Use the “differentiation by first principles” formula.

b

Function is $f(x) = x^2$

Chapter 12 · Original slide 8 · Test Your Understanding

Prove from first principles that the derivative of x^4 is $4x^3$.

As $h \to 0$, any term involving a multiplication by h will become 0.

Support:

“Row 4” of Pascal’s Triangle is:

1 4 6 4 1

You’re welcome.

Chapter 12 · Original slide 9 · Exercise 12B – Pearson Pure Y1 page 261-262

5 Prove, from first principles, that the derivative of $6x$ is 6.

Reminder:

$f\prime(x)$ and $\frac{dy}{dx}$ both mean the gradient function, also known as the derivative of y.

$f'(x) = \frac{dy}{dx} = \lim_{h \to 0} \frac{f(x+h)-f(x)}{h}$

Chapter 12 · Original slide 10 · Exercise 12B – Pearson Pure Y1 page 261-262

6 Prove, from first principles, that the derivative of $4x^2$ is $8x$.

Reminder:

$f\prime(x)$ and $\frac{dy}{dx}$ both mean the gradient function, also known as the derivative of y.

$f'(x) = \frac{dy}{dx} = \lim_{h \to 0} \frac{f(x+h)-f(x)}{h}$

Chapter 12 · Original slide 11 · Exercise 12B – Pearson Pure Y1 page 261-262

7 $\mathrm{f}(x) = ax^2$, where a is a constant. Prove, from first principles, that $\mathrm{f}'(x) = 2ax$.

Reminder:

$f\prime(x)$ and $\frac{dy}{dx}$ both mean the gradient function, also known as the derivative of y.

$f'(x) = \frac{dy}{dx} = \lim_{h\to 0} \frac{f(x+h)-f(x)}{h}$

Chapter 12 · Original slide 12 · Exercise 12B – Pearson Pure Y1 page 261-262

Challenge $\mathrm{f}(x) = \frac{1}{x}$.

a Given that $\mathrm{f}'(x) = \lim_{h\to 0} \frac{\mathrm{f}(x+h)-\mathrm{f}(x)}{h}$, show that $\mathrm{f}'(x) = \lim_{h\to 0} \frac{-1}{x^2+xh}$.

b Deduce that $\mathrm{f}'(x) = -\frac{1}{x^2}$.

Chapter 12: Differentiation

Chapter 12 · Original slide 13 · Just for your interest…

Why couldn't we just immediately make h equal to 0 in $\lim_{h\to 0} \frac{(x+h)^2-x^2}{h}$?

$\frac{0}{0}$ is known as an indeterminate form. Whereas we know what happens with expressions like $\frac{1}{0}$ (i.e. its value tends towards infinity), indeterminate forms are bad because their values are ambiguous, and prevent $\lim$ expressions from being evaluated.

Consider $\frac{0}{0}$ for example: 0 divided by anything usually gives 0, but anything divided by 0 is usually infinity. We can see these conflict.

If we just stick $h = 0$ in straight away:

$\lim_{h\to 0} \frac{(x+0)^2-x^2}{0} = \lim_{h\to 0} \frac{0}{0}$

Wait, uh oh...

Thankfully, when indeterminate forms appear in $\lim$ expressions, there are variety of techniques to turn the expression into one that doesn't have any indeterminate forms.

One simple technique, that worked in our example, is to expand and simplify. This gave us $\lim_{h\to 0} (2x + h)$ and $2x + 0$ clearly is fine. Other techniques are more advanced.

This is important when sketching harder functions: http://www.drfrostmaths.com/resources/resource.php?rid=163

Can you guess some other indeterminate forms, i.e. expressions whose value is ambiguous?

Chapter 12 · Original slide 14 · Differentiating {x}^{n}

Thankfully, there's a quick way to differentiate terms of the form x^n (where n is a constant) with having to use first principles every time:

! If $y = ax^n$ then $\frac{dy}{dx} = nax^{n-1}$ (where a, n are constants)

i.e. multiply by the power and reduce the power by 1

Examples:

$y = x^5$

Power is 5, so multiply by 5 then reduce power by 5.

The power need not be an integer!

Remember to use $f'(x)$ not $\frac{dy}{dx}$

$f(x) = x^{\frac{1}{2}}$

$y = 2x^6$

Why would it be incorrect to say that $\mathbf{y=2^x}$ differentiates to $\mathbf{\frac{dy}{dx}=x2^{x-1}}$?

The rule only works when the base is x and the power is a constant. Neither is true here! Note that x^n is "a power of x" whereas 2^x is an exponential term (which you will encounter more in Chp14), and therefore differentiate differently. You will learn how to differentiate exponential terms in Year 2.

$f(x) = \frac{x}{x^4}$

$y = \sqrt{x^6}$

Chapter 12: Differentiation

Chapter 12 · Original slide 15 · Test Your Understanding

1

$y = x^7$

2

$y = 3x^{10}$

3

$f(x) = \frac{x^{\frac{1}{2}}}{x^2}$

4

$y = ax^a$

5

$f(x) = \sqrt{49x^7}$

Chapter 12 · Original slide 16 · Differentiating Multiple Terms

Differentiate $y = x^2 + 4x + 3$

First thing to note:

If $y = f(x) + g(x)$ then $\frac{dy}{dx} = f'(x) + g'(x)$

i.e. differentiate each term individually in a sum/ subtraction.

$y = 4x = 4x^1$

Therefore applying the usual rule:

$\frac{dy}{dx} = 4x^0 = 4$

Alternatively, if you compare $y = 4x$ to $y = mx + c$, it's clear that the gradient is fixed and $m = 4$.

$y = 3 = 3x^0$

Therefore applying the usual rule:

$\frac{dy}{dx} = 0x^{-1} = 0$

Alternatively, if you sketch $y = 4$, the line is horizontal, so the gradient is 0.

Chapter 12 · Original slide 17 · Quickfire Questions

1

$y = 2x^2 - 3x$

2

$y = 4 - 9x^3$

3

$y = 5x + 1$

(where a is a constant)

4

$y = ax$

(where p is a constant)

5

$y = 6x - 3 + px^2$

Chapter 12 · Original slide 18 · Harder Example

Let $f(x) = 4x^2 - 8x + 3$

a) Find the gradient of $y = f(x)$ at the point $\left(\frac{1}{2}, 0\right)$

b) Find the coordinates of the point on the graph of $y = f(x)$ where the gradient is 8.

c) Find the gradient of $y = f(x)$ at the points where the curve meets the line $y = 4x - 5$.

Remember that the 'gradient function' allows you to find the gradient for a particular value of x.

b

This example is important!

Previously you used a value of x to get the gradient $f'(x)$. This time we're doing the opposite: using a known gradient $f'(x)$ to get the value of x. We therefore substitute $f'(x)$ for 8.

c

Once you have your x, you need to work out y.

Ensure you use the correct equation!

Chapter 12 · Original slide 19 · Test Your Understanding

Let $f(x) = x^2 - 4x + 2$

a) Find the gradient of $y = f(x)$ at the point $(1, -1)$

b) Find the coordinates of the point on the graph of $y = f(x)$ where the gradient is 5.

c) Find the gradient of $y = f(x)$ at the points where the curve meets the line $y = 2 - x$.

Chapter 12 · Original slide 20 · Exercise 12D – Pearson Pure Y1 page 265-266

5 Find the gradients of the curve $y = x^2 - 5x + 10$ at the points A and B where the curve meets the line $y = 4$.

Chapter 12 · Original slide 21 · Exercise 12D – Pearson Pure Y1 page 265-266

6 Find the gradients of the curve $y = 2x^2$ at the points C and D where the curve meets the line $y = x + 3$.

Chapter 12 · Original slide 22 · Exercise 12D – Pearson Pure Y1 page 265-266

7 $\mathrm{f}(x) = x^2 - 2x - 8$.
a Sketch the graph of $y = \mathrm{f}(x)$.
b On the same set of axes, sketch the graph of $y = \mathrm{f}'(x)$.
c Explain why the x-coordinate of the turning point of $y = \mathrm{f}(x)$ is the same as the x-coordinate of the point where the graph of $y = \mathrm{f}'(x)$ crosses the x-axis.

Chapter 12 · Original slide 23 · Differentiating Harder Expressions

If your expression isn't a sum of x^n terms, simply manipulate it until it is!

$y = \sqrt{x}$

2. Split up fractions.

3. Expand out brackets.

NOT $3x^{-1}$!!!

$y = \frac{1}{3x}$

1. Turn roots into powers:

$y = \frac{1}{\sqrt[3]{x}}$

$y = \frac{x^2+3}{\sqrt{x}}$

$y = x^2\,(x-3)$

4. Beware of numbers in denominators!

Chapter 12 · Original slide 24 · Test Your Understanding

Differentiate the following.

$y = \frac{2+x^3}{x^2}$

$y = \frac{1}{\sqrt{x}}$

$y = \frac{1+2x}{3x\sqrt{x}}$

Chapter 12 · Original slide 25 · Exercise 12E – Pearson Pure Y1 page 267-268

6 $\mathrm{f}(x) = \frac{12}{p\sqrt{x}} + x$, where p is a real constant and $x > 0$. Given that $\mathrm{f}'(2) = 3$, find p, giving your answer in the form $a\sqrt{2}$ where a is a rational number.

Chapter 12 · Original slide 26 · Exercise 12E – Pearson Pure Y1 page 267-268

7 $\mathrm{f}(x) = (2 - x)^9$.

a Find the first 3 terms, in ascending powers of x, of the binomial expansion of $\mathrm{f}(x)$, giving each term in its simplest form.

b If x is small, so that x^2 and higher powers can be ignored, show that $\mathrm{f}'(x) \approx 9216x - 2304$.

Chapter 12 · Original slide 27 · Extension

[MAT 2013 1E]

The expression
$\frac{d^2}{dx^2}[(2x-1)^4(1-x)^5] + \frac{d}{dx}[(2x+1)^4(3x^2-2)^2]$
is a polynomial of degree:

9

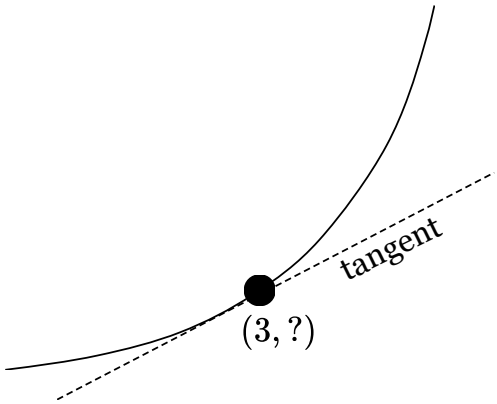
8

7

less than 7

This just means “differentiate twice”. The ‘second derivative’ comes later in this chapter.

Chapter 12 · Original slide 28 · Finding equations of tangents



Find the equation of the tangent to the curve $y = x^2$ when $x = 3$.

We want to use $y - y_1 = m\,(x - x_1)$ for the tangent (as it is a straight line!). Therefore we need:

(a) A point (x_1, y_1)

(b) The gradient m.

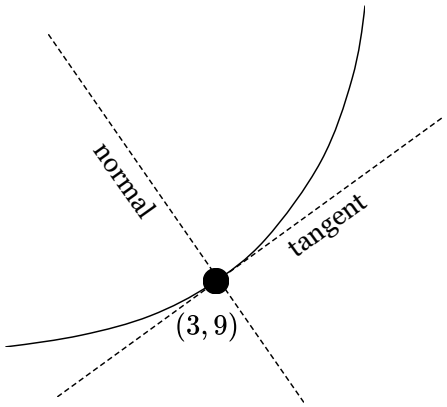
Gradient function:

Gradient when $x = 3$:

y-value when $x = 3$:

So equation of tangent:

Chapter 12 · Original slide 29 · Finding equations of normals



Find the equation of the normal to the curve $y = x^2$ when $x = 3$.

The normal to a curve is the line perpendicular to the tangent.

Tip: A very common error is for students to accidentally forget whether the question is asking for the tangent or for the normal.

Equation of tangent (from earlier):

Therefore equation of normal:

Chapter 12 · Original slide 30 · Test Your Understanding

Find the equation of the normal to the curve $y = x + 3\sqrt{x}$ when $x = 9$.

Tip: I like to use m_T and m_N to make clear to the examiner (and myself) what gradient I've found.

Chapter 12 · Original slide 31 · Exercise 12F – Pearson Pure Y1 page 269-270

3 Find the coordinates of the point where the tangent to the curve $y = x^2 + 1$ at the point $(2, 5)$ meets the normal to the same curve at the point $(1, 2)$.

Chapter 12 · Original slide 32 · Exercise 12F – Pearson Pure Y1 page 269-270

4 Find the equations of the normals to the curve $y = x + x^3$ at the points $(0, 0)$ and $(1, 2)$, and find the coordinates of the point where these normals meet.

Chapter 12 · Original slide 33 · Exercise 12F – Pearson Pure Y1 page 269-270

5 For $\mathrm{f}(x) = 12 - 4x + 2x^2$, find the equations of the tangent and the normal at the point where $x = -1$ on the curve with equation $y = \mathrm{f}(x)$.

Chapter 12 · Original slide 34 · Exercise 12F – Pearson Pure Y1 page 269-270

6 The point P with x-coordinate $\frac{1}{2}$ lies on the curve with equation $y = 2x^2$. The normal to the curve at P intersects the curve at points P and Q. Find the coordinates of Q. (6 marks)

Chapter 12: Differentiation

Chapter 12 · Original slide 35 · Extension

The 'difference of two cubes':

$a^3 - b^3 = (a - b)\left(a^2 + ab + b^2\right)$

will help for both of these.

1

[STEP I 2005 Q2]

The point P has coordinates $\left(p^2, 2p\right)$ and the point Q has coordinates $\left(q^2, 2q\right)$, where p and q are non-zero and $p \neq q$. The curve C is given by $y^2 = 4x$. The point R is the intersection of the tangent to C at P and the tangent to C at Q. Show that R has coordinates $(pq, p + q)$.

The point S is the intersection of the normal to C at P and the normal to C at Q. If p and q are such that $(1, 0)$ lies on the line PQ, show that S has coordinates $\left(p^2 + q^2 + 1, p + q\right)$, and that the quadrilateral $PSQR$ is a rectangle.

Solutions on next slide.

2

[STEP I 2012 Q4]

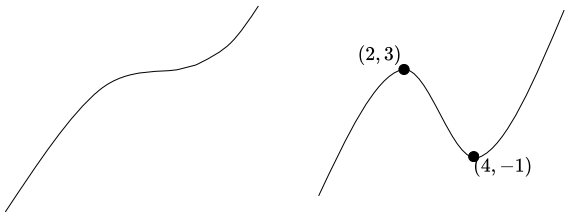
The curve C has equation $xy = \frac{1}{2}$.

The tangents to C at the distinct points $P\left(p, \frac{1}{2p}\right)$ and $Q\left(q, \frac{1}{2q}\right)$, where p and q are positive, intersect at T and the normal to C at these points intersect at N. Show that T is the point $\left(\frac{2pq}{p+q}, \frac{1}{p+q}\right)$

In the case $pq = \frac{1}{2}$, find the coordinates of N. Show (in this case) that T and N lie on the line $y = x$ and are such that the product of their distances from the origin is constant.

Chapter 12 · Original slide 36 · Solutions to Extension Question 1

Chapter 12 · Original slide 37 · Increasing and Decreasing Functions



A function can also be increasing and decreasing in certain intervals.

What do you think it means for a function to be an ‘increasing function’?

! An increasing function is one whose gradient is always at least 0.

$f'(x) \geq 0$ for all x.

We could also write “$f(x)$ is decreasing in the interval $[2, 4]$”

[a,b] represents all the real numbers between **a** and **b** inclusive, i.e:

$[a, b] = \{x : a \leq x \leq b\}$

It would be ‘strictly increasing’ if $f(x) > 0$ for all x, i.e. is not allowed to go horizontal.

Chapter 12 · Original slide 38 · Examples

Show that the function $f(x) = x^3 + 6x^2 + 21x + 2$ is increasing for all real values of x.

Tip: To show a quadratic is always positive, complete the square, then indicate the squared term is always at least 0.

Find the interval on which the function $f(x) = x^3 + 3x^2 - 9x$ is decreasing.

Chapter 12 · Original slide 39 · Test Your Understanding

Show that the function $f(x) = x^3 + 16x - 2$ is increasing for all real values of x.

Find the interval on which the function $f(x) = x^3 + 6x^2 - 135x$ is decreasing.

Chapter 12 · Original slide 40 · Exercise 12G – Pearson Pure Y1 page 271

3 Show that the function $\mathrm{f}(x) = 4 - x(2x^2 + 3)$ is decreasing for all $x \in \mathbb{R}$.

Chapter 12 · Original slide 41 · Exercise 12G – Pearson Pure Y1 page 271

4 a Given that the function $\mathrm{f}(x) = x^2 + px$ is increasing on the interval $[-1, 1]$, find one possible value for p.
b State with justification whether this is the only possible value for p.

Chapter 12 · Original slide 42 · Second Order Derivatives

When you differentiate once, the expression you get is known as the first derivative.

Unsurprisingly, when we differentiate a second time, the resulting expression is known as the second derivative. And so on…

Original Function	
First Derivative	Second Derivative
Leibniz's	$\frac{dy}{dx} = 4x^3$
$\frac{d^2y}{dx^2} = 12x^2$	$y = x^4$
$y\prime = 4x^3$	$y\prime\prime = 12x^2$
Newton's	$\dot{y} = 4x^3$
$\ddot{y} = 12x^2$	$f(x) = x^4$
Lagrange's	$f\prime(x) = 4x^3$
$f\prime\prime(x) = 12x^2$	You can similarly have the third derivative ($\frac{d^3y}{dx^3}$), although this is no longer in the A Level syllabus. We'll see why might use the second derivative soon…

Chapter 12 · Original slide 43 · Just for your interest…

How does the notation $\frac{d^2y}{dx^2}$ work?

Why are the squareds where they are?

The $\frac{d}{dx}(\dots)$ notation is quite handy, because it behaves as a function and allows us to write the original expression and the derivative within a single equation:

$$\frac{d}{dx}\left(x^3+1\right)=3x^2$$

Suppose that $y=x^3+1$.

Then when we write $\frac{dy}{dx}$, we're effectively doing $\frac{d(x^3+1)}{dx}$ (by substitution), although this would typically be written:

$$\frac{d}{dx}\left(x^3+1\right)$$

Therefore, if we wanted to differentiate y twice, we'd do:

$$\begin{aligned}&\frac{d}{dx}\left(\frac{d}{dx}(y)\right)\\&=\frac{d^2}{dx^2}(y)\\&=\frac{d^2y}{dx^2}\end{aligned}$$

Chapter 12 · Original slide 44 · Examples

If $y=3x^5+\frac{4}{x^2}$, find $\frac{d^2y}{dx^2}$.

If $f(x)=3\sqrt{x}+\frac{1}{2\sqrt{x}}$, find $f''(x)$.

Chapter 12: Differentiation

Chapter 12 · Original slide 45 · Test Your Understanding

If $y = 5x^3 - \frac{x}{3\sqrt{x}}$, find $\frac{d^2y}{dx^2}$.

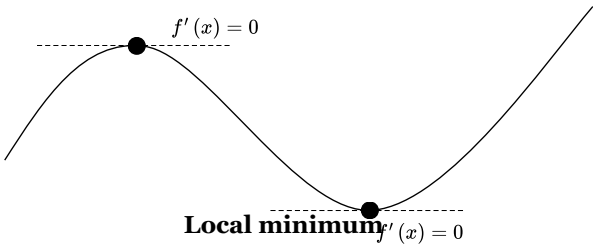
Chapter 12 · Original slide 46 · Exercise 12H – Pearson Pure Y1 page 272

3 Given that $y = (2x - 3)^3$, find the value of x when $\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2} = 0$.

Chapter 12 · Original slide 47 · Exercise 12H – Pearson Pure Y1 page 272

4 $\mathrm{f}(x) = px^3 - 3px^2 + x^2 - 4$. When $x = 2$, $\mathrm{f}''(x) = -1$. Find the value of p.

Chapter 12 · Original slide 48 · Stationary/Turning Points



A stationary point is where the gradient is 0, i.e. $f'(x) = 0$.

Local maximum

Note: It's called a 'local' maximum because it's the function's largest output within the vicinity. Functions may also have a 'global' maximum, i.e. the maximum output across the entire function. This particular function doesn't have a global maximum because the output keeps increasing up to infinity. It similarly has no global minimum, as with all cubics.

Find the coordinates of the turning points of $y = x^3 + 6x^2 - 135x$

Chapter 12 · Original slide 49 · More Examples

Find the least value of
$f(x) = x^2 - 4x + 9$

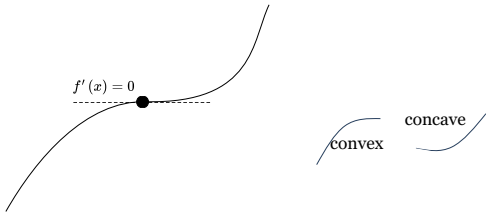
Method 1: Differentiation

Note: Method 2 is only applicable for quadratic functions. For others, differentiation must be used.

Find the turning point of
$y = \sqrt{x} - x$

Method 2: Completing the square

Chapter 12 · Original slide 50 · Points of Inflection



There's a third type of stationary point (that we've encountered previously):

Side Note: Not all points of inflection are stationary points, as can be seen in the example on the right.
A point of inflection which **is** a stationary point is known as a *saddle point*.



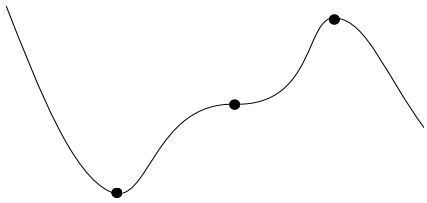
A point of inflection is where the curve changes from convex concave (or vice versa).

(the same terms used in optics!)

Technically we could label these either way round depending on where we view the curve from. What's important is that the concavity changes.

i.e. the line curves in one direction before the point of inflection, then curves in the other direction after.

Chapter 12 · Original slide 51 · How do we tell what type of stationary point?



Method 1: Look at gradient just before and just after point.

Local Maximum		
Gradient just before	Gradient at maximum	Gradient just after

Point of Inflection		
Gradient just before	Gradient at p.o.i	Gradient just after

Local Minimum		
Gradient just before	Gradient at minimum	Gradient just after

Chapter 12 · Original slide 52 · How do we tell what type of stationary point?

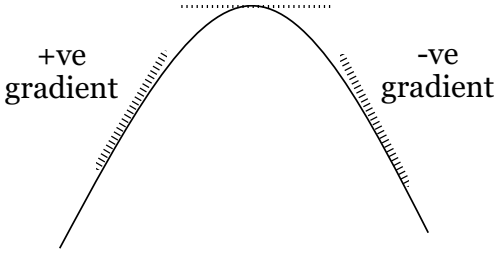
Method 1: Look at gradient just before and just after point.

Strategy: Find the gradient for values just before and after $x = 2$. Let's try $x = 1.9$ and $x = 2.1$.

Find the stationary point on the curve with equation $y = x^4 - 32x$, and determine whether it is a local maximum, a local minimum or a point of inflection.

	x=1.9	**x=2**	**x=2.1**
Gradient			
Shape			

Chapter 12 · Original slide 53 · Method 2: Using the second derivative



The method of substituting values of x just before and after is a bit cumbersome.

It also has the potential for problems: what if two different types of stationary points are really close together?

Recall the gradient gives a measure of the rate of change of y, i.e. how much the y value changes as x changes.

Thus by differentiating the gradient function, the second derivative tells us the rate at what the gradient is changing.

Thus if the second derivative is positive, the gradient is increasing.

If the second derivative is negative, the gradient is decreasing.

At a maximum point, we can see that as x increases, the gradient is decreasing from a positive value to a negative value.

$\therefore \frac{d^2y}{dx^2} < 0$

0

gradient

Chapter 12 · Original slide 54 · Method 2: Using the second derivative

! At a stationary point $x = a$:

• If $f''(a) > 0$ the point is a local minimum.

• If $f''(a) < 0$ the point is a local maximum.

• If $f''(a) = 0$ it could be any type of point, so resort to Method 1.

I will eventually do a 'Just for your Interest...' thingy on why we can't classify the point when $f''(x) = 0$, and how we could use the third derivative!

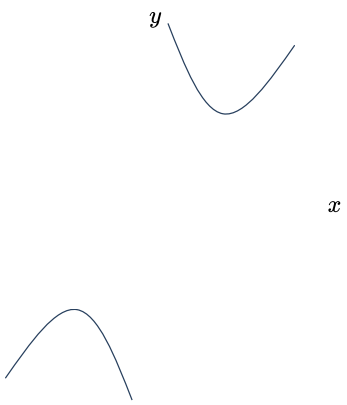
The stationary point of $y = x^4 - 32x$ is $(2, -48)$. Use the second derivative to classify this stationary point.

Find $\frac{d^2y}{dx^2}$ for the x at the stationary point.

Chapter 12 · Original slide 55 · Test Your Understanding

The curve with equation $y = x^2 - 32\sqrt{x} + 20$, $x > 0$, has a stationary point P. Use calculus
(a) to find the coordinates of P, (6)
(b) to determine the nature of the stationary point P. (3)

Chapter 12 · Original slide 56 · Sketching Graphs



All the way back in Chapter 4, we used features such as intercepts with the axes, and behaviour when $x \to \infty$ and $x \to -\infty$ in order to sketch graphs.

Now we can also find stationary/turning points!

[Textbook] By first finding the stationary points, sketch the graph of $y = \frac{1}{x} + 27x^3$

? As $x \to \infty$, $x \to -\infty$

Chapter 12 · Original slide 57 · Exercise 12I – Pearson Pure Y1 page 276

5 By considering the gradient on either side of the stationary point on the curve $y = x^3 - 3x^2 + 3x$, show that this point is a point of inflection. Sketch the curve $y = x^3 - 3x^2 + 3x$.

Chapter 12 · Original slide 58 · Exercise 12I – Pearson Pure Y1 page 276

6 Find the maximum value and hence the range of values for the function $\mathrm{f}(x) = 27 - 2x^4$.

Chapter 12 · Original slide 59 · Exercise 12I – Pearson Pure Y1 page 276

7 $f(x) = x^4 + 3x^3 - 5x^2 - 3x + 1$.
a Find the coordinates of the stationary points of $f(x)$, and determine the nature of each.
b Sketch the graph of $y = f(x)$.

Chapter 12 · Original slide 60 · Extension

1
[MAT 2014 1C] The cubic
$y = kx^3 - (k+1)x^2 + (2-k)x - k$
has a turning point, that is a minimum, when $x = 1$ precisely for
A) $k > 0$
B) $0 < k < 1$
C) $k > \frac{1}{2}$
D) $k < 3$
E) all values of k

3
[MAT 2001 1E] The maximum gradient of the curve $y = x^4 - 4x^3 + 4x^2 + 2$ in the range $0 \le x \le 2\frac{1}{5}$ occurs when:
A) $x = 0$
B) $x = 1 - \frac{1}{\sqrt{3}}$
C) $x = 1 + \frac{1}{\sqrt{3}}$
D) $x = 2\frac{1}{5}$
[STEP I 2007 Q8] A curve is given by:
$y = ax^3 - 6ax^2 + (12a + 12)x - (8a + 16)$
where a is a real number. Show that this curve touches the curve with equation $y = x^3$ at $(2, 8)$. Determine the coordinates of any other point of intersection of the two curves.
(i) Sketch on the same axes these two curves when $a = 2$.
(ii) ... when $a = 1$ (iii) when $a = -2$

4
Hint: When two curves touch, their y values must match, but what else must also match?

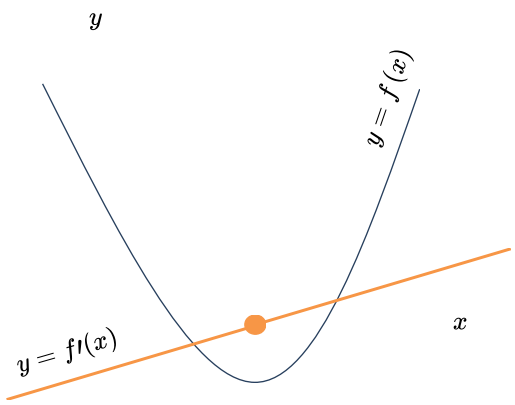
2
[MAT 2004 1B] The smallest value of the function:
$f(x) = 2x^3 - 9x^2 + 12x + 3$
In the range $0 \le x \le 2$ is what?

Chapter 12 · Original slide 61 · Solutions to Extension Questions

Chapter 12 · Original slide 62 · Solutions to Extension Questions

Chapter 12 · Original slide 63 · Solutions to Extension Questions

Chapter 12 · Original slide 64 · Sketching Gradient Functions



The new A Level specification specifically mentions being able to sketch $y = f\prime(x)$.

If you know the function $f\prime(x)$ explicitly (e.g. because you differentiated $y = f(x)$), you can use your knowledge of sketching straight line/ quadratic/cubic graphs.

But in other cases you won't be given the function explicitly, but just the sketch.

The gradient of $y = f(x)$ is negative, but increasing.

The gradient at the turning point is 0.

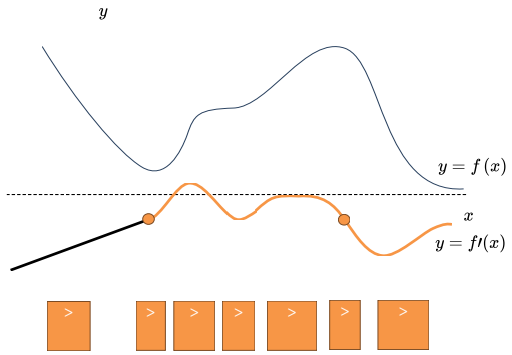
The gradient of $y = f(x)$ is positive, and increasing.

Click to Sketch $y = f\prime(x)$

Click to Sketch $y = f\prime(x)$

Click to Sketch $y = f\prime(x)$

Chapter 12 · Original slide 65 · A Harder One



Tip: Mentally describe the gradient of each section of the curve, starting your mental sentence with “The gradient is...” and using terms like “positive”, “but increasing”, “but not changing”, etc.

Gradient is negative, but increasing.

Gradient is positive: initially increases but then decreases.

Gradient is positive: initially increases but then decreases.

Gradient is 0.

Gradient is 0, but positive before and after.

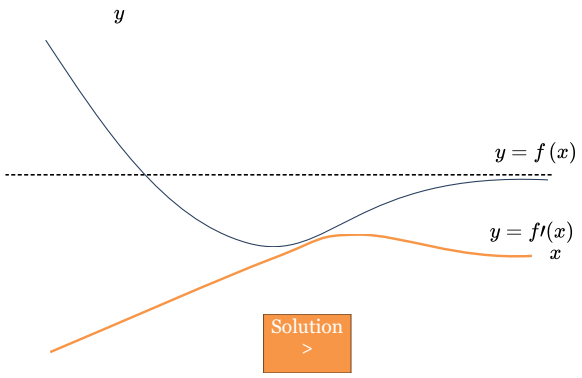
Gradient is 0.

Gradient is negative. Initially decreases but then tends towards 0.

Tip: The gradient is momentarily not changing so the gradient of the gradient is 0. We get a turning point in $y = f'(x)$ whenever the curve has a point of inflection.

Again, on $y = f(x)$ this is a point of inflection so the gradient function has a turning point. It is also a stationary point, so $f'(x) = 0$. i.e. The $f\prime(x)$ curve touches the x-axis.

Chapter 12 · Original slide 66 · Test Your Understanding



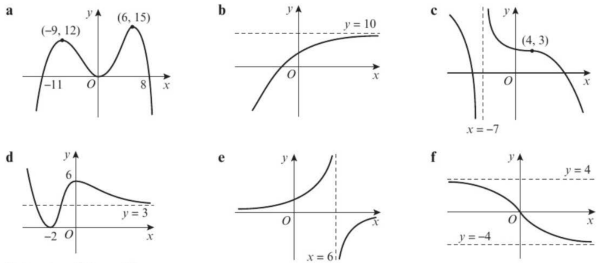
Gradient is negative, but increasing.

Gradient is positive; initially increases but then decreases.

Gradient is 0.

Gradient is positive, but tending towards 0.

Chapter 12 · Original slide 67 · Exercise 12J – Pearson Pure Y1 page 278



1 For each graph given, sketch the graph of the corresponding gradient function on a separate set of axes. Show the coordinates of any points where the curve cuts or meets the x-axis, and give the equations of any asymptotes.

Solutions here

Chapter 12 · Original slide 68 · Exercise 12J – Pearson Pure Y1 page 278

2 $\mathrm{f}(x) = (x+1)(x-4)^2$.
a Sketch the graph of $y = \mathrm{f}(x)$.
b On a separate set of axes, sketch the graph of $y = \mathrm{f}'(x)$.
c Show that $\mathrm{f}'(x) = (x-4)(3x-2)$.
d Use the derivative to determine the exact coordinates of the points where the gradient function cuts the coordinate axes.

Chapter 12 · Original slide 69 · Extension

1

[MAT 2015 1B]

$f(x) = (x + a)^n$

where a is a real number and n is a positive whole number, and $n \geq 2$. If $y = f(x)$ and $y = f\prime(x)$ are plotted on the same axes, the number of intersections between $f(x)$ and $f'(x)$ will:

A) always be odd

B) always be even

C) depend on a but not n

D) depend on n but not a

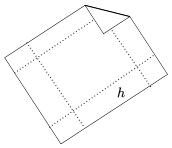
E) depend on both a and n.

Recall that:

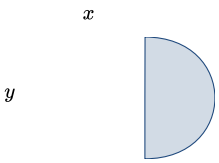
Odd function: $f(-x) = -f(x)$

Even function: $f(-x) = f(x)$

Chapter 12 · Original slide 70 · Optimisation Problems/Modelling



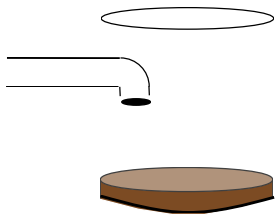
We have a sheet of A4 paper, which we want to fold into a cuboid. What height should we choose for the cuboid in order to maximise the volume?



These are examples of optimisation problems: we're trying to maximise/minimise some quantity by choosing an appropriate value of a variable that we can control.

We have 50m of fencing, and want to make a bear pen of the following shape, such that the area is maximised. What should we choose x and y to be?

Chapter 12 · Original slide 71 · ‘Rate of change’



Up to now we’ve had y in terms of x, where $\frac{dy}{dx}$ means “the rate at which y changes with respect to x”.

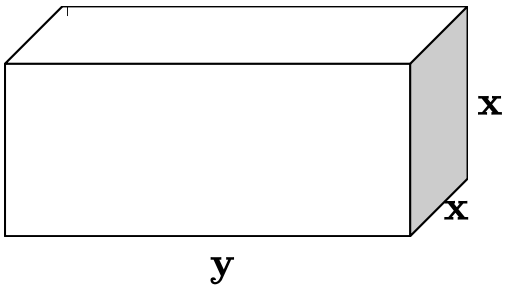
But we can use similar gradient function notation for other physical quantities.

Real Life Maths!

A sewage container fills at a rate of 20 cm3 per second.

How could we use appropriate notation to represent this?

Chapter 12 · Original slide 72 · Example Optimisation Problem



Optimisation problems in an exam usually follow the following pattern:

• There are 2 variables involved (you may have to introduce one yourself), typically lengths.

• There are expressions for two different physical quantities:

• One is a constraint, e.g. “the surface area is 20cm2”.

• The other we wish to maximise/minimise, e.g. “we wish to maximise the volume”.

• We use the constraint to eliminate one of the variables in the latter equation, so that it is then just in terms of one variable, and we can then use differentiation to find the turning point.

[Textbook] A large tank in the shape of a cuboid is to be made from 54m2 of sheet metal. The tank has a horizontal base and no top. The height of the tank is x metres. Two of the opposite vertical faces are squares.

a) Show that the volume, V m3, of the tank is given by $V = 18x - \frac{2}{3}x^3$.

These are the two equations mentioned in the guidance: one for surface area and one volume.

We need to introduce a second variable ourselves so that we can find expressions for the surface area and volume.

Once we have the ‘optimal’ value of x, we sub it back into V to get the best possible volume.

b) Given that x can vary, use differentiation to find the maximum or minimum value of V.

Chapter 12 · Original slide 73 · Test Your Understanding

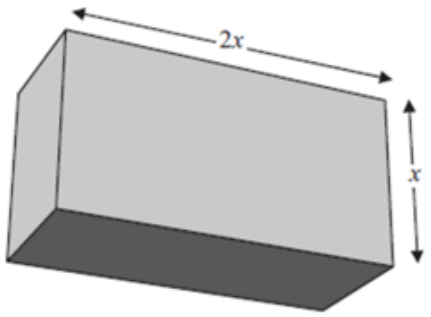


Figure 2

A cuboid has a rectangular cross-section where the length of the rectangle is equal to twice its width, x cm, as shown in Figure 2.

The volume of the cuboid is 81 cubic centimetres.

(a) Show that the total length, L cm, of the twelve edges of the cuboid is given by $L = 12x + \frac{162}{x^2}$.

(3)

(b) Use calculus to find the minimum value of L. (6)

(c) Justify, by further differentiation, that the value of L that you have found is a minimum. (2)

Chapter 12 · Original slide 74 · Exercise 12K – Pearson Pure Y1 page 81

6 A rectangular garden is fenced on three sides, and the house forms the fourth side of the rectangle.

a Given that the total length of the fence is 80 m, show that the area, A, of the garden is given by the formula $A = y(80 - 2y)$, where y is the distance from the house to the end of the garden.

b Given that the area is a maximum for this length of fence, find the dimensions of the enclosed garden, and the area which is enclosed.

Chapter 12 · Original slide 75 · Exercise 12K – Pearson Pure Y1 page 81

7 A closed cylinder has total surface area equal to 600π.
a Show that the volume, $V\,\text{cm}^3$, of this cylinder is given by the formula $V = 300\pi r - \pi r^3$, where $r\,\text{cm}$ is the radius of the cylinder.
b Find the maximum volume of such a cylinder.

Chapter 12 · Original slide 76 · Exercise 12K – Pearson Pure Y1 page 81

9 A shape consists of a rectangular base with a semicircular top, as shown.
a Given that the perimeter of the shape is 40 cm, show that its area, A cm^2, is given by the formula
$A = 40r - 2r^2 - \frac{\pi r^2}{2}$
where r cm is the radius of the semicircle. (2 ma
b Hence find the maximum value for the area of the shape. (4 ma



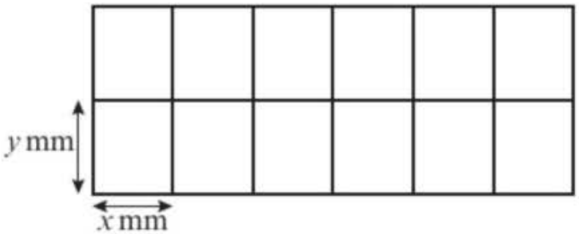
9 A shape consists of a rectangular base with a semicircular top, as shown.

a Given that the perimeter of the shape is $40\,\text{cm}$, show that its area, $A\,\text{cm}^2$, is given by the formula $A = 40r - 2r^2 - \frac{\pi r^2}{2}$

where $r\,\text{cm}$ is the radius of the semicircle. (2 marks)

b Hence find the maximum value for the area of the shape. (4 marks)

Chapter 12 · Original slide 77 · Exercise 12K – Pearson Pure Y1 page 81



10 The shape shown is a wire frame in the form of a large rectangle split by parallel lengths of wire into 12 smaller equal-sized rectangles.

a Given that the total length of wire used to complete the whole frame is $1512\,\text{mm}$, show that the area of the whole shape, $A\,\text{mm}^2$, is given by the formula

$$A = 1296x - \frac{108x^2}{7}$$

where $x\,\text{mm}$ is the width of one of the smaller rectangles. (4 marks)

b Hence find the maximum area which can be enclosed in this way. (4 marks)

Chapter 12 · Original slide 78 · Extension

1

[STEP I 2006 Q2] A small goat is tethered by a rope to a point at ground level on a side of a square barn which stands in a large horizontal field of grass. The sides of the barn are of length $2a$ and the rope is of length $4a$. Let A be the area of the grass that the goat can graze.

Prove that $A \leq 14\pi a^2$ and determine the minimum value of A.